Alpha decay 3

In this section we derive a regression formula for the following half-life data. Rows with $Z = 90$ are uranium and $Z = 88$ are thorium. Recall that Z is atomic number minus two.

$\log \tau_{1/2}$	Z	E
6.30	90	6.80
14.4	90	5.99
21.5	90	5.41
29.7	90	4.86
34.2	90	4.57
39.5	90	4.27
7.52	88	6.45
17.9	88	5.52
28.5	88	4.77
40.6	88	4.08

From the previous sections we have

$$\log \tau_{1/2} \propto \gamma \qquad \gamma = C_1 \frac{Z}{\sqrt{E}} - C_2 \sqrt{Z r_1}$$

Noting that $r_1 \propto A^{1/3}$ and to a good approximation $A \propto Z$ we have

$$\sqrt{Z r_1} \propto \sqrt{Z Z^{1/3}} = Z^{2/3}$$

Hence the Geiger–Nuttall regression model

$$\log \tau_{1/2} = \beta_2 \frac{Z}{\sqrt{E}} + \beta_1 Z^{2/3} + \beta_0$$

By ordinary least squares the best coefficients for the above data are

$$\begin{aligned} \beta_2 &= 3.702 \\ \beta_1 &= -3.104 \\ \beta_0 &= -59.28 \end{aligned}$$

The regression formula yields the following predicted values.

$\log \tau_{1/2}$	Predicted $\log \tau_{1/2}$
6.30	6.15
14.4	14.5
21.5	21.6
29.7	29.6
34.2	34.2
39.5	39.6
7.52	7.56
17.9	18.0
28.5	28.5
40.6	40.6

The observed linearity of $\log \tau_{1/2}$ and $Z/\sqrt{E}$ confirms the tunneling hypothesis for alpha decay.

Eigenmath script